

Lecture 5: Logistic Regression

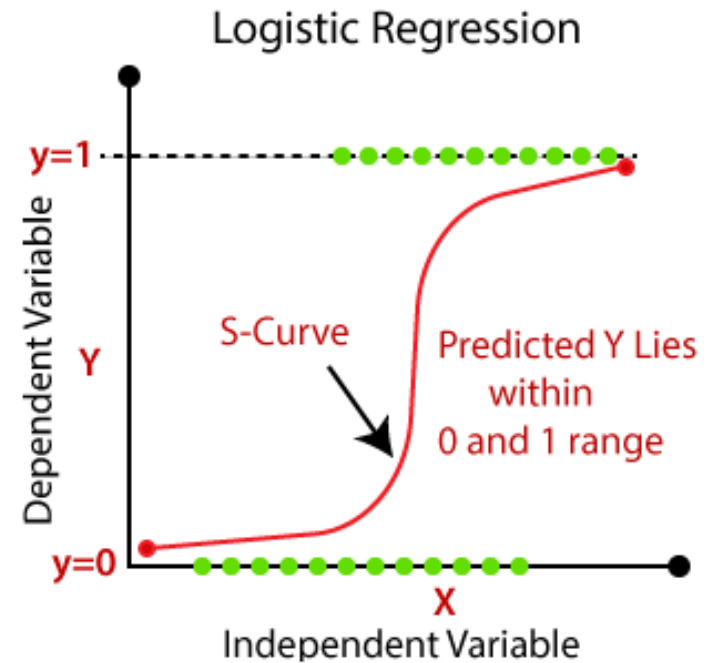
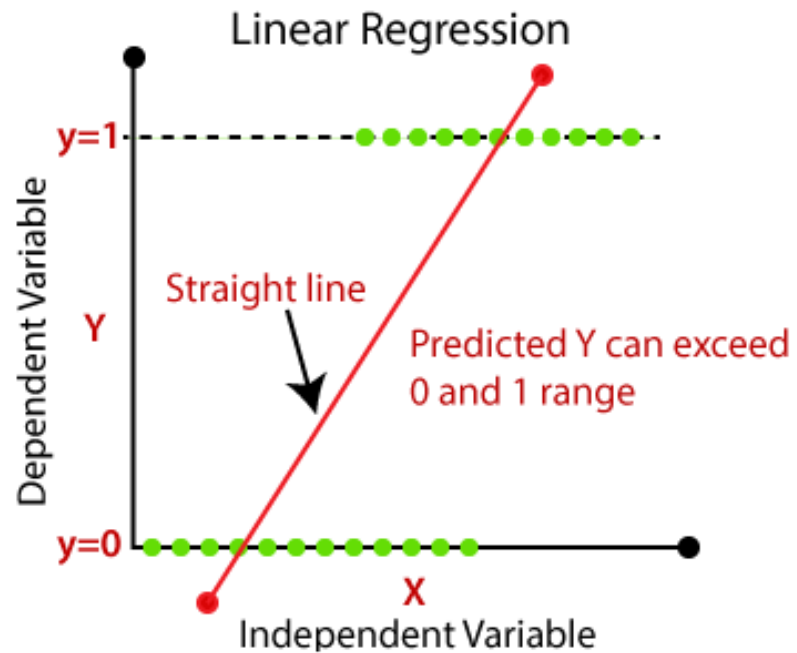
ML/AI for Engineering Applications
ENV 6932

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Outline

- Logistic Regression
- Bias-Variance Tradeoff
- Regularization
- Ridge Regression
- Lasso Regression

Logistic Regression

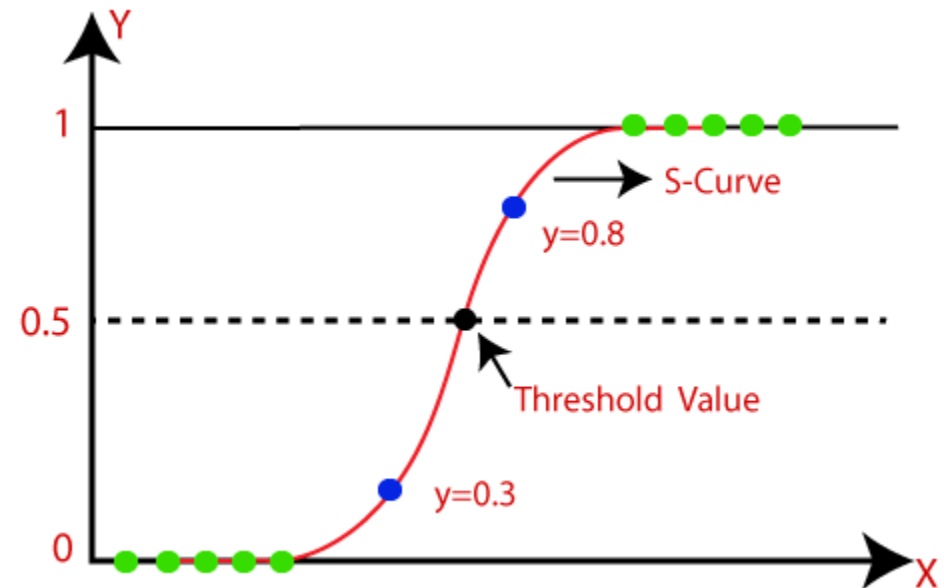


Logistic Regression

- One of the most popular ML algorithm.
- Used for Classification as well as Regression, but mainly Classification.
- The output of Binary Logistic Regression can be only between the 0 and 1.
- **Example:** whether it will rain today or not, 0 or 1, true or false.

Logistic Regression

- Logistic regression is based on the concept of Maximum Likelihood estimation. According to this estimation, the observed data should be most probable.
- The weighted sum of inputs is fed through an activation function that can map values in between 0 and 1.
- Such activation function is known as **sigmoid function**, sigmoid curve or S-curve.



Logistic Regression

- Independent variables can be numeric or categorical variables, but the dependent variable will always be categorical
- Logistic regression is a statistical model that uses Logistic function to model the conditional probability.
- Calculate the conditional probability of the dependent variable Y, given independent variable X:

$$P(Y=1|X)$$

$$P(Y=0|X)$$

Logistic Regression

- $P(Y | X)$ is approximated as a sigmoid function applied to a linear combination of input features.
- If we predict the probability via linear regression, we can state it as:

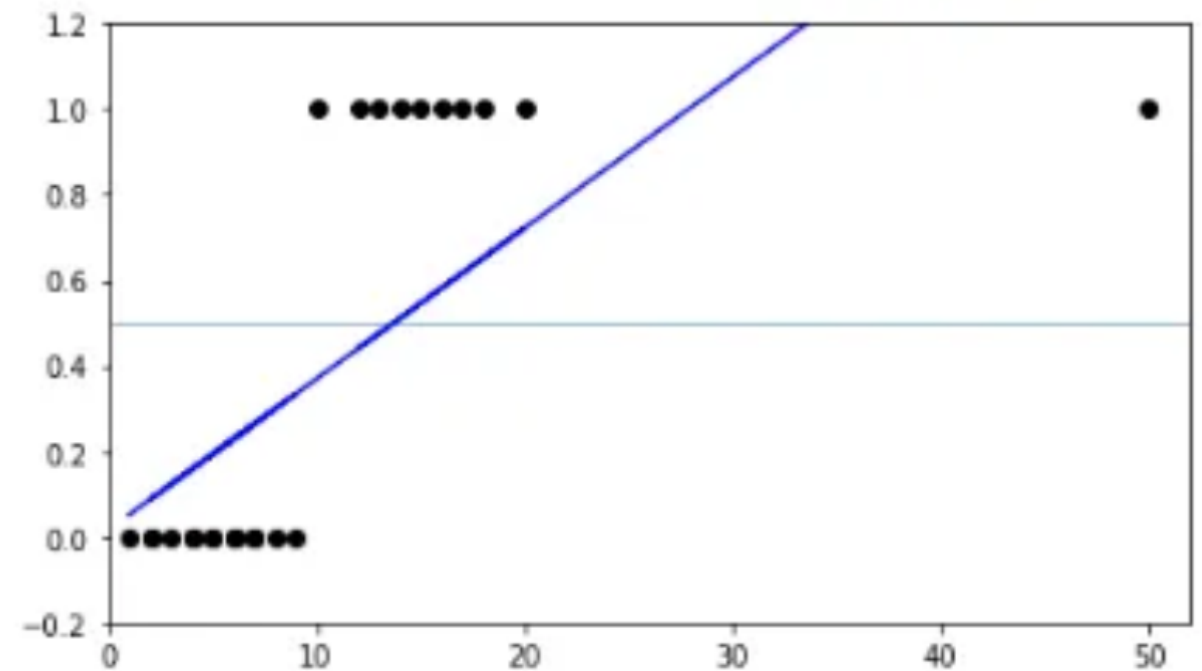
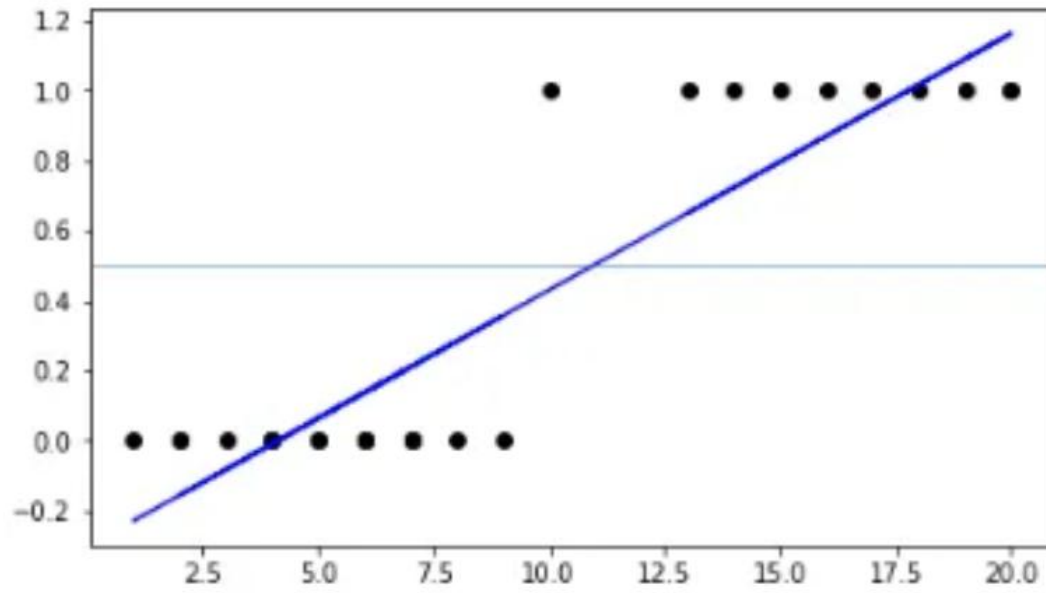
$$p(X) = \beta_0 + \beta_1 X,$$

$$\text{where } p(X) = P(y = 1 | X)$$

- Linear regression model can generate the ***predicted probability*** as any number ranging from negative to positive infinity, whereas probability of an outcome can only lie between $0 < P(x) < 1$.

Logistic Regression

- Also, Linear regression has a considerable effect on outliers.



Logit Function

- where, the left hand side is called the **logit** or log-odds function, and **$p(x)/(1-p(x))$** is called odds.

$$\log \left(\frac{p(X)}{1 - p(X)} \right) = \beta_0 + \beta_1 X$$

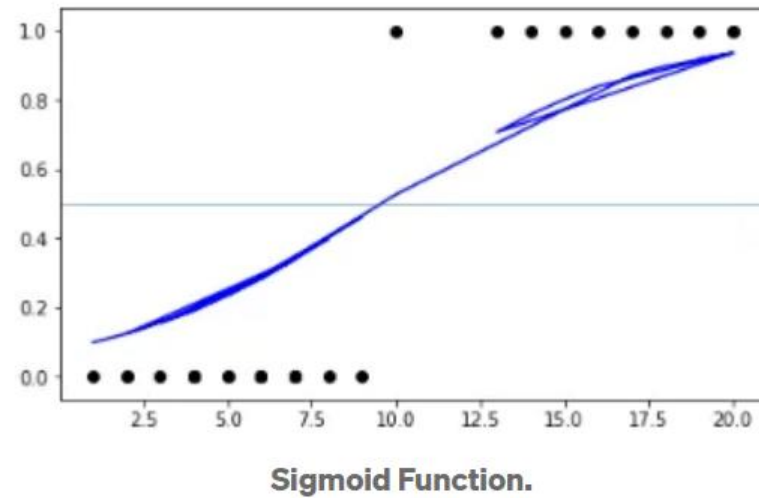
- The *odds* signifies the ratio of probability of success to probability of failure. Therefore, in Logistic Regression, linear combination of inputs are mapped to the log(odds) - the output being equal to 1.

- If we take an **inverse of the above function**, we get:

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

- This is known as the *Sigmoid function* and it gives an S-shaped curve. It always gives a value of probability ranging from $0 < p < 1$.

- If we take an **inverse** of the above function, we get:



The Logistic Regression Model

Binary outcome (Y) variable

$$y_i \sim \text{Binom}(pi)$$

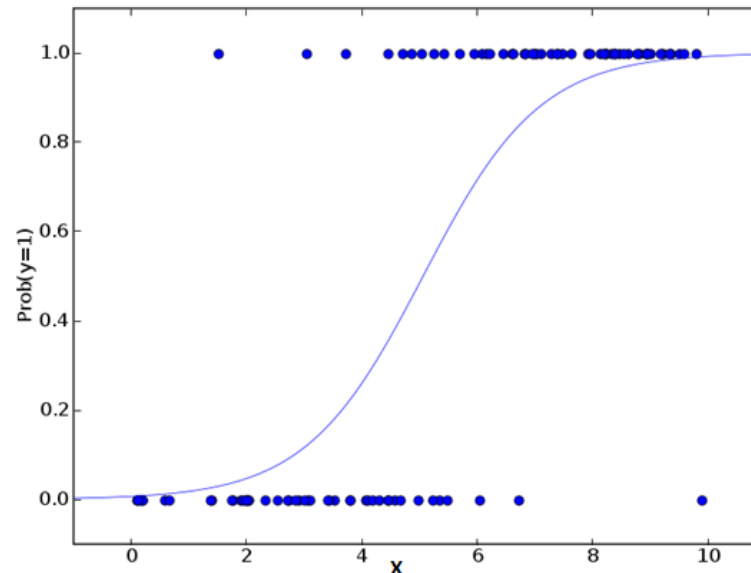
Number of successes

$$\text{Logit}(p) = \log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n$$

$$p_i = \frac{\exp(\beta_0 + \beta_1 x_{1,i} + \cdots + \beta_n x_{n,i})}{1 + \exp(\beta_0 + \beta_1 x_{1,i} + \cdots + \beta_n x_{n,i})}$$

Logistic Regression

- Logistic regression is used when the dependent variable is binary (0/ 1, True/ False, Yes/ No).
- The value of Y ranges from 0 to 1 and it can be represented by following equation:
- $\text{Logit}(p) = \ln(p/(1 - p)) = B_0 + B_1x_1 + B_2x_2 + B_3x_3 \dots + B_kx_k$



Formula

$$P_x = \binom{n}{x} p^x q^{n-x}$$

P = binomial probability

x = number of times for a specific outcome within n trials

$\binom{n}{x}$ = number of combinations

p = probability of success on a single trial

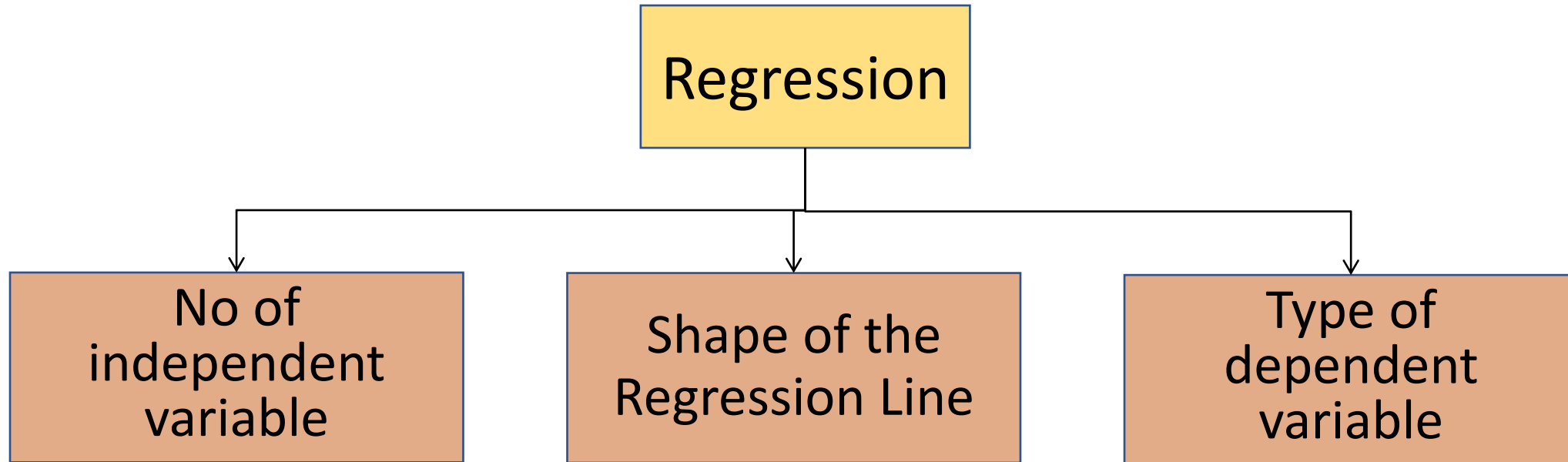
q = probability of failure on a single trial

n = number of trials

Types of Regressions

- Linear Regression
- Logistic Regression
- Polynomial Regression
- Stepwise Regression
- Ridge Regression
- Lasso Regression
- ElasticNet Regression

How to determine the proper regression type?

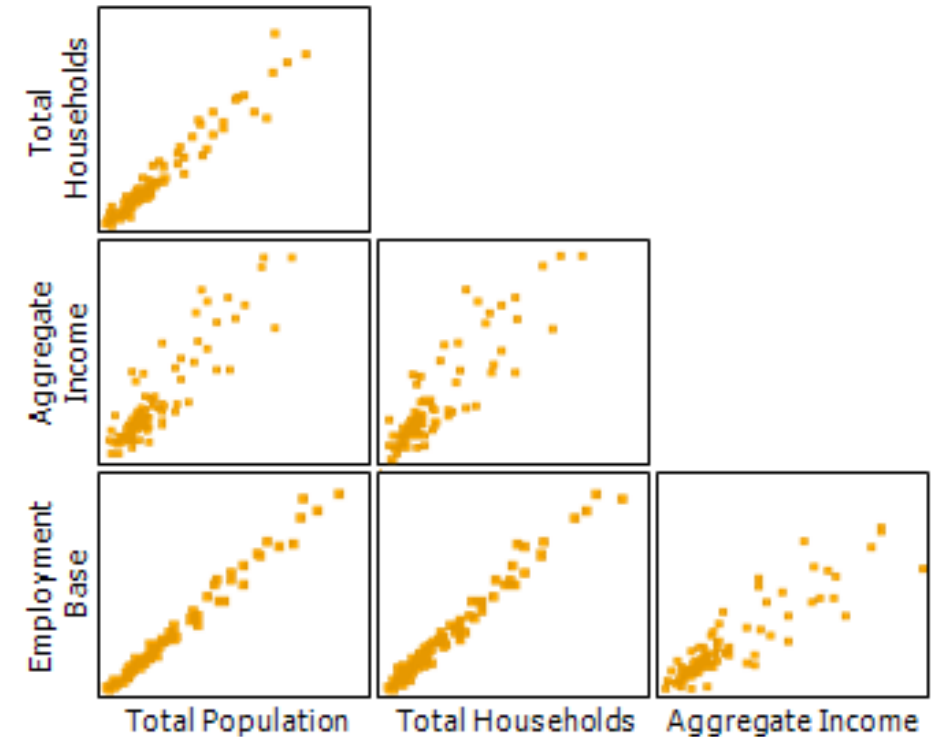


Important Points: Linear Regression

- There must be **linear relationship** between predictors and dependent variables
- Multiple regression suffers from **multicollinearity, autocorrelation, heteroskedasticity**.

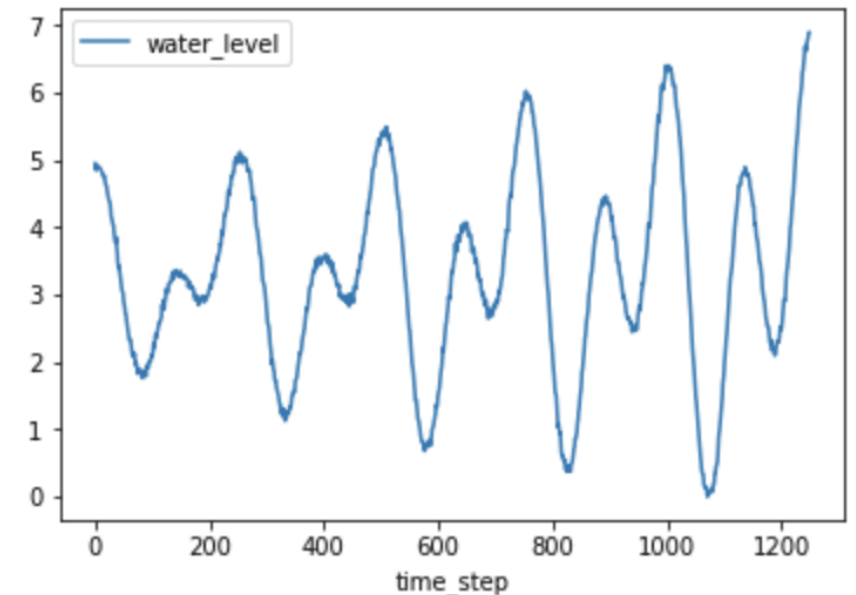
Important Points: Linear Regression

- **Multicollinearity:** one predictor variable can be linearly predicted from the others with a substantial degree of accuracy.
- **Solution:**
- Create a correlation matrix in order to check for multicollinearity.
- Drop the highly correlated variables
- Use principal component analysis, which can help to reduce the number of variables and increase independent terms.



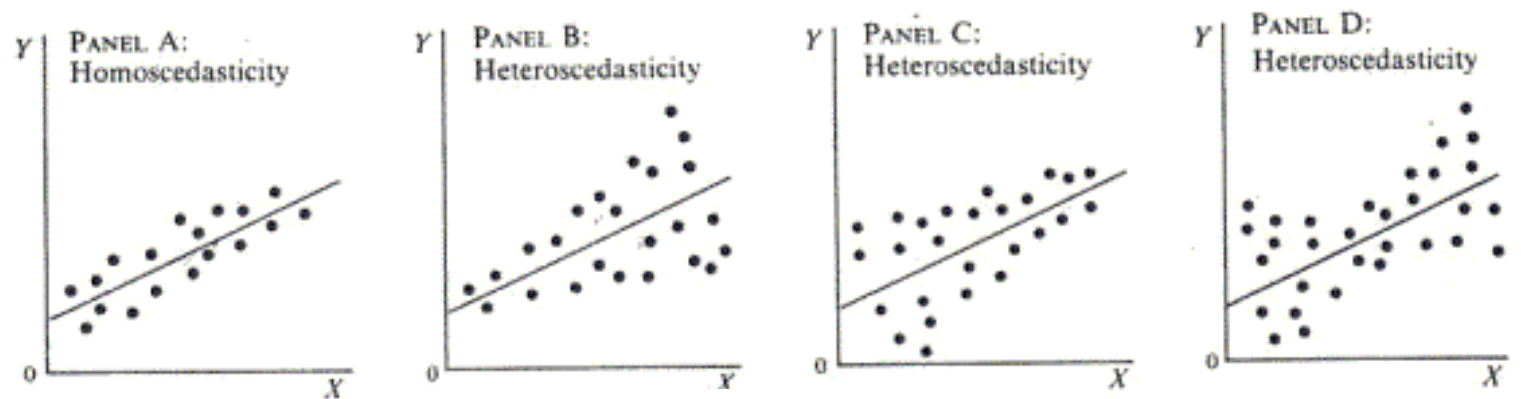
Important Points: Linear Regression

- **Autocorrelation:** or serial correlation, is the correlation of a signal with a delayed copy of itself as a function of delay. There is nonzero correlation between variables at different time point.
- **Solution:** remove seasonality



Important Points: Linear Regression

- **Heteroskedasticity:** means not all of the random variables have the same variance, situations where the variance of the residuals is unequal over a range of measured values
- **Solution:** Re-build the model with new predictors.
Variable transformation (e.g. Box-Cox transformation)



Important Points: Linear Regression

- **Multicollinearity:** one predictor variable can be linearly predicted from the others with a substantial degree of accuracy.
- **Autocorrelation:** or serial correlation, is the correlation of a signal with a delayed copy of itself as a function of delay. There is nonzero correlation between variables at different time point.
- **Heteroskedasticity:** means not all of the random variables have the same variance, situations where the variance of the residuals is unequal over a range of measured values

Important Points: Linear Regression

- Linear Regression is very sensitive to **Outliers**.
- Multicollinearity can increase the variance of the coefficient estimates and make the estimates very sensitive to minor changes in the model.
- In case of multiple independent variables, most significant independent variables can be selected with **forward selection, backward elimination** and **step wise approach**.

Important Points: Logistic Regression

- Widely used for **classification problems**
- Logistic regression doesn't require linear relationship, it can handle various types of relationships as it uses a non-linear log transformation.
- To avoid over fitting and under fitting, include all significant variables. A good approach is to use a step wise method.

Important Points: Logistic Regression

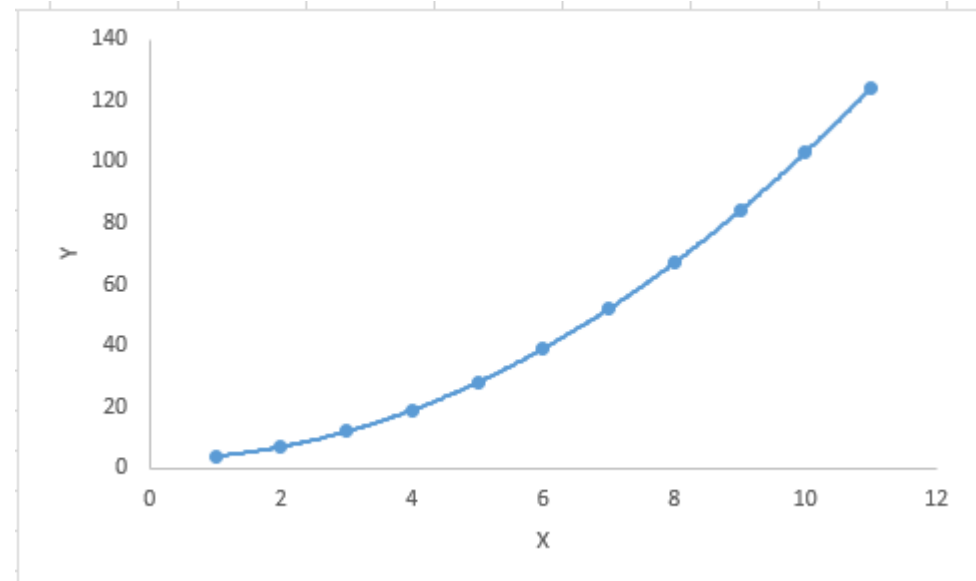
- It needs **large sample sizes** as maximum likelihood estimates are less powerful at low sample sizes than ordinary least square.
- The independent variables should not be correlated with each other i.e. **no multicollinearity**. However, interaction effects of categorical variables can be included in the analysis and in the model.

Important Points: Logistic Regression

- **Ordinal logistic regression:** If the values of dependent variable is ordinal.
- **Multinomial Logistic regression:** If dependent variable is multi class then it is known as Multinomial Logistic Regression.

Polynomial Regression

- Polynomial regression: if the power of independent variable is more than 1.



References

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